COLLEGE OF CREATIVE INNOVATION

Programme

Master of Software Engineering
(Level 9) 180 Credits

Course

MSE804: Blockchain and decentralised digital identity
(15 Credits)

Assessment 2

Case Study with Presentation

Weighting within the course:

50%

Assessment Tasks to Learning Outcome and GPOs mapping

Tasks	Learning Outcomes	GPOs
Task 1	LO2	GPO3
Task 2	LO3	GPO2
Task 3	LO4	GPO1

Objective

The aim of this assessment is to critically evaluate blockchain technology and decentralised digital identity principles, emphasising their application in meeting specific software industry requirements. Additionally, the assessment seeks to analyse the associated challenges, risks, regulations, prospects, and best practices in integrating decentralised digital identity and blockchain within software design solutions. Furthermore, it aims to integrate blockchain technology, cryptography, decentralised systems architectures, and information systems to develop innovative and robust systems and services. Ultimately, the objective is to demonstrate a comprehensive understanding of these technologies, addressing industry demands and advancing the field of software design.

Assessment Instructions

- This assessment is an open book activity, you can use your own course and review notes as well as offline or online resources, such as textbooks or online journals.
- You can always ask your tutor if you need further explanation or if the instructions are not clear
- By completing and submitting an assessment you are authenticating that you are the original creator and/or author of all the submitted work and that it does not violate plagiarism or copyright law. All written work such as essays, reflections, reports must be in your own words. Please refer to the Academic Misconduct and Authenticity of Assessment Evidence statement in your Student Handbook for more information.
- Please ensure the completion of the assessment by **week 15.**
- Grades and feedback will be returned within 15 days of the submission date.

Learning Outcomes

LO2: Analyse the challenges, risks, regulations, prospects, and best practices associated with decentralised digital identity and blockchain usage in software design solutions.

LO3: Integrate blockchain technology, cryptography, decentralised systems architectures, and information systems into innovative systems and services.

LO4: Critically assess the application of blockchain technology to software engineering systems.

Graduate Profile Outcomes (GPOs) covered

GPO1: Develop advanced software engineering knowledge and skills and apply these to solve emerging or existing problems.

GPO2: Utilise highly specialised knowledge and skills to carry out cutting-edge software engineering projects independently and collaboratively.

GPO3: Develop and apply professional and ethical standards in software engineering to meet the industry's expectations and the ability to work with integrity in compliance with organisational criteria.

Success Criteria: You need to meet all the requirements of each of the learning outcomes and receive 50% or more to pass this assessment. You are allowed a maximum of three attempts. To meet all the requirements of each of the learning outcomes, you must achieve PASS results for each task item.

Grading: The final grade will be determined by the score achieved in this assessment based on the following table. Should a second or third attempt be required the maximum contribution toward the overall mark for the tasks that required a second or third assessment attempt is 50%. A late submission is considered a second attempt, so the contribution will be capped at 50%.

Grade	Mark Band Range
A+	Meet all course requirements, mark range (90-100)
A	Meet all course requirements, mark range (85-89)
A-	Meet all course requirements, mark range (80-84)
B+	Meet all course requirements, mark range (75-79)
B	Meet all course requirements, mark range (70-74)
B-	Meet all course requirements, mark range (65-69)
C+	Meet all course requirements, mark range (60-64)
C	Meet all course requirements, mark range (55-59)
C-	Meet all course requirements, mark range (50-54)
D	Did not meet all course requirements, mark range (40-49)
E	Did not meet all course requirements, mark range (0-39)

Submission requirements:

- Submit a Word or PDF document that contains your answers to all tasks below.
- Ensure the submission of your assessment to **Turnitin** and generate a report adhering to the acceptable threshold of 20% or lower for originality.
- Download the files from Turnitin and subsequently submit them to Blackboard. Include the Turnitin report as part of your final submission file.
- This is to be submitted by the due date.
- The submission must have your name and ID number printed.
- Submission should be typed, double-spaced, and adhere to a standard citation format if external sources are referenced.

Assessment Tasks

Case Study: Blockchain Technology for Food Traceability in Retail Stores

Your task is to prepare a short report addressing the questions outlined below for each learning outcome. Your report should demonstrate your understanding of blockchain technology and its potential applications in ensuring food safety and traceability in retail stores. You should also provide recommendations for how Countdown can effectively integrate blockchain technology into its existing systems to improve food traceability.

Introduction:

In the modern food industry, ensuring the safety and traceability of food products is crucial. Traditional methods of tracking food products through the supply chain, such as paper-based and digital data systems, are often inefficient and error-prone. Blockchain technology offers a promising solution to address these challenges effectively. In this case study, you will explore how blockchain technology can benefit retail food stores, using Countdown as a focal point.

Case Scenario:

John, the CEO of Countdown, is interested in leveraging blockchain technology to enhance the efficiency of supply chain management and ensure food safety and traceability. He tasks the Chief Technical Officer, Luke, to arrange a short report and a presentation on blockchain technology for supply chain management, specifically focusing on its potential applications for Countdown. Luke assigns the task to the team, requiring a short report explaining various aspects of blockchain technology in the context of food traceability.

Task 1: Challenges and Best Practices

LO2: Analyse the challenges, risks, regulations, prospects, and best practices associated with decentralised digital identity and blockchain usage in software design solutions.

- What challenges might Countdown face in implementing blockchain technology for supply chain management, particularly regarding decentralised digital identity and data management?
- What regulatory frameworks and compliance measures should Countdown consider when implementing blockchain technology to ensure data security and privacy?
- What are the best practices for Countdown to consider blockchain as a solution for supply chain management system, and what are the potential long-term benefits or prospects of using blockchain technology in this context?

Task 2: Integrating Blockchain

LO3: Integrate blockchain technology, cryptography, decentralised systems architectures, and information systems into innovative systems and services.

- How can blockchain technology be integrated into Countdown's supply chain management system to enhance traceability and transparency?
- What role does cryptography play in securing the data stored on the blockchain for Countdown's supply chain management system?
- How can a decentralised systems architecture be designed for Countdown's supply chain management system to ensure scalability and efficiency?

Task 3: Assessing Blockchain

LO4: Critically assess the application of blockchain technology to software engineering systems.

- Critically evaluate how blockchain technology could enhance efficiency and traceability in Countdown's supply chain management system compared to traditional methods.
- Assess the benefits and drawbacks of applying blockchain technology to software engineering systems in the context of food traceability and analyse how these factors impact the decision-making process for implementation.
- Submit your presentation on the critical assessment of the application of blockchain technology to software engineering systems.

Marking Rubric

Task 1: Challenges and Best Practices

LO2: Analyse the challenges, risks, regulations, prospects, and best practices associated with decentralised digital identity and blockchain usage in software design solutions.

Criterion & Weighting		**A (80-100) %**	**B (65-79) %**	**C (50-64) %**	**D (40-49) %**	**E (0-39) %**
Challenges and Risks Analysis (Decentralized Digital Identity & Data Management)	15%	Demonstrated a comprehensive understanding of challenges related to: - Decentralized identity management: user adoption, verification, and potential security vulnerabilities. - Data management: data quality, access control, and potential privacy concerns	Identified key challenges related to decentralized identity and data management. May have missed some specific details or nuances.	Identified some challenges related to decentralized identity and data management. Analysis lacked depth or specificity.	Limited understanding of challenges related to decentralized identity and data management.	Demonstrated inadequate understanding of the challenges associated with decentralized identity and data management, failing to grasp implications for security and privacy in distributed systems
Regulatory Framework and Compliance Analysis	10%	Demonstrated thorough research and understanding of relevant regulatory frameworks for data security and privacy in the context of blockchain technology. Identified specific compliance measures Countdown should consider.	Presented a good overview of relevant regulations and compliance considerations. May not have provided detailed examples or specific measures.	Identified some relevant regulations but lacked in-depth understanding or specific recommendations.	Limited understanding of relevant regulations and compliance requirements.	Showed minimal awareness of relevant regulations and compliance requirements, neglecting crucial legal frameworks essential for ensuring data protection and organizational adherence
Best Practices and Long-Term Benefits Analysis	10%	Provided well-developed and detailed best practices, tailored to Countdown's specific needs, and explained their potential benefits.	Identified some best practices but lacked a comprehensive or well-organized discussion. May have discussed some long-term benefits but lacked depth or clarity.	Mentioned some best practices or benefits but lacked detailed explanation or analysis.	Limited understanding of best practices or long-term potential of blockchain in supply chain management.	Exhibited limited awareness of blockchain's best practices and its potential long-term benefits in enhancing transparency and efficiency within supply chain management

Task 2: Integrating Blockchain

LO3: Integrate blockchain technology, cryptography, decentralised systems architectures, and information systems into innovative systems and services.

Criterion & Weighting		**A (80-100) %**	**B (65-79) %**	**C (50-64) %**	**D (40-49) %**	**E (0-39) %**
Blockchain Integration	10%	Demonstrated a clear and comprehensive understanding of how to integrate blockchain into Countdown's supply chain, providing specific examples of data points and functionalities.	Explained the integration of blockchain with relevant details but may lack specific examples.	Showed basic understanding of integrating blockchain but lacked clarity or depth.	Limited understanding of blockchain integration or irrelevant examples.	Demonstrated insufficient understanding of blockchain integration, providing irrelevant examples that did not align with practical applications or industry standards
Cryptography	10%	Explained the role of cryptography in securing data and transactions on the blockchain in detail, providing specific examples of cryptographic techniques used.	Mentioned the role of cryptography but lacks specific details or explanations of its applications.	Briefly mentioned cryptography but lacks clarity or understanding of its purpose.	Limited or incorrect understanding of the role of cryptography.	Exhibited limited or incorrect understanding of the fundamental role of cryptography, failing to explain its significance in securing digital transactions and data.
Decentralised Systems Architecture	10%	Described a well-designed and detailed decentralised system architecture for Countdown's supply chain, addressing scalability and efficiency considerations.	Presented a basic understanding of a decentralised architecture but may lack specific details or justifications for design choices.	Mentioned a decentralized architecture but lacked clarity or explanation of its components and functionalities.	Limited or irrelevant understanding of a decentralized architecture.	Showed inadequate or irrelevant understanding of decentralized architecture, lacking insights into its operational principles and implications for scalability and resilience

Task 3: Assessing Blockchain

LO4: Critically assess the application of blockchain technology to software engineering systems.

Criterion & Weighting		**A (80-100) %**	**B (65-79) %**	**C (50-64) %**	**D (40-49) %**	**E (0-39) %**
Critical Evaluation	10%	Provided a well-balanced and insightful evaluation of how blockchain could enhance efficiency and traceability compared to traditional methods, considering specific examples relevant to Countdown's supply chain.	Provided a clear evaluation of the benefits and drawbacks but may lack detailed analysis or specific examples.	Presented a basic understanding of the benefits and drawbacks but lacked depth or critical analysis.	Limited understanding of the potential impact of blockchain or irrelevant comparisons.	Demonstrated a deficient understanding of the potential impact of blockchain, often presenting irrelevant comparisons that did not contribute to a meaningful assessment
Benefit and Drawback Analysis	10%	Demonstrated a comprehensive understanding of the benefits and drawbacks of using blockchain for food traceability, providing detailed explanations and relevant examples.	Mentioned the benefits and drawbacks but lacked specific details or explanations of their implications.	Briefly mentioned benefits and drawbacks but lacked clarity or understanding of their significance.	Limited understanding of the benefits and drawbacks or irrelevant information.	Evidenced a shallow grasp of blockchain's benefits and drawbacks, often including irrelevant information
Decision-Making Factors	10%	Presented a well-structured and insightful analysis of the factors influencing the decision to implement blockchain in Countdown's context, justifying the importance of each factor.	Discussed factors influencing the decision but lacked a clear structure or detailed reasoning for their importance.	Mentioned some factors but lacked a clear analysis or justification for their impact on the decision.	Limited understanding of the factors influencing the decision or irrelevant arguments.	Showed inadequate grasp of the factors influencing decisions, presenting irrelevant arguments.
Presentation	5%	The presentation was exceptionally well-organized, clear, and engaging. All points were thoroughly researched, and the arguments were compelling. Visual aids were used effectively to	The presentation was well-organized and mostly clear. Most points were well-researched, and the arguments were logical. Visual aids were used adequately. Demonstrated a good	The presentation was somewhat organized and clear. Some points were adequately researched, but there were gaps in the arguments. Visual aids were used, but their effectiveness was limited.	The presentation was disorganized and unclear. Few points were adequately researched, and the arguments were weak. Visual aids were minimal or ineffective. Demonstrated a limited	The presentation lacked organization and clarity, with poorly researched points and weak arguments. Visual aids were minimal or ineffective, failing to enhance understanding.

		enhance understanding. Demonstrated a deep understanding of the topic and addressed all aspects of the assessment criteria. Recommendations were insightful and supported by evidence.	understanding of the topic and addressed most aspects of the assessment criteria. Recommendations were reasonable and supported by some evidence.	Demonstrated a basic understanding of the topic and addressed some aspects of the assessment criteria. Recommendations were provided but lacked depth and evidence.	understanding of the topic and did not address most aspects of the assessment criteria. Recommendations were lacking or not supported by evidence.	There was a limited grasp of the topic, neglecting key aspects of the assessment criteria, and recommendations were either unsupported or absent of evidence

Note: *The ranges for each grade level encompass the full 11-point grading system as outlined in the accompanying table. Please refer to the table for detailed percentage ranges associated with each letter grade.*